

Practical File
On
DATABASE MANAGEMENT SYSTEMS

Submitted in partial fulfilment of the requirements for
the award of the degree

Bachelor of Technology
in
Department of Computer Science and Engineering



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Ex – 1

Employee Directory

1. Aim

To design a relational schema for an employee directory with departments and employees, enforce primary-key and foreign-key constraints, create a reporting view, and verify the design with sample queries in SQLite.

2. Theory

2.1 Tables and keys

A departments table stores each organisational unit. An employees table stores people; department_id is a foreign key to departments, and manager_id is a self-referencing foreign key for the reporting hierarchy.

2.2 Views

A view (employee_directory) joins employees to their department and manager names so reporting queries do not repeat join logic.

3. Schema

Table	Key columns	Role
departments	department_id PK	Organisational units
employees	employee_id PK; department_id FK	Staff records
employees	manager_id FK → employees	Reporting chain
employee_directory	VIEW	Flattened directory for queries

4. Procedure

1. Create departments and employees with constraints and indexes.
2. Create the employee_directory view.
3. Insert six departments and twenty-six employees with a three-tier org chart.
4. Run aggregate queries on the view (counts by department).

5. Source Code

File: 19-08-2026/schema.sql

```
-- =====
-- Employee Directory – schema
-- SQLite dialect. Runs as-is in the in-browser SQL Compiler
-- (index.html), and in the sqlite3 CLI / DB Browser for SQLite.
-- =====

PRAGMA foreign_keys = ON;

DROP VIEW IF EXISTS employee_directory;
DROP TABLE IF EXISTS employees;
DROP TABLE IF EXISTS departments;

-- -----
-- departments
-- -----
CREATE TABLE departments (
    department_id INTEGER PRIMARY KEY AUTOINCREMENT,
    name          TEXT NOT NULL UNIQUE,
```

```

        description    TEXT,
        color_hex      TEXT NOT NULL DEFAULT '#1CB0F6'
    );

-- -----
-- employees
-- -----
CREATE TABLE employees (
    employee_id        INTEGER PRIMARY KEY AUTOINCREMENT,
    first_name         TEXT NOT NULL,
    last_name          TEXT NOT NULL,
    email              TEXT NOT NULL UNIQUE,
    job_title           TEXT NOT NULL,
    department_id       INTEGER NOT NULL REFERENCES departments (department_id) ON DELETE RESTRICT,
    manager_id         INTEGER REFERENCES employees (employee_id) ON DELETE SET NULL,
    salary              NUMERIC(10,2) NOT NULL CHECK (salary > 0),
    hire_date           TEXT NOT NULL,
    employment_status  TEXT NOT NULL DEFAULT 'active'
                    CHECK (employment_status IN ('active', 'remote', 'on_leave')),
    avatar_seed        TEXT NOT NULL
);

CREATE INDEX idx_employees_department ON employees (department_id);
CREATE INDEX idx_employees_manager   ON employees (manager_id);

-- -----
-- employee_directory – flattened view joining department + manager.
-- Convenient for reporting queries and reused by the frontend.
-- -----
CREATE VIEW employee_directory AS
SELECT
    e.employee_id,
    e.first_name || ' ' || e.last_name AS full_name,
    e.email,
    e.job_title,
    d.name                               AS department_name,
    d.color_hex                         AS department_color,
    m.first_name || ' ' || m.last_name AS manager_name,
    e.salary,
    e.hire_date,
    e.employment_status,
    e.avatar_seed
FROM employees e
JOIN departments d ON d.department_id = e.department_id
LEFT JOIN employees m ON m.employee_id = e.manager_id;

```

6. Output

```
SQL> SELECT COUNT(*) AS dept_count FROM departments;
dept_count
6
```

```
SQL> SELECT COUNT(*) AS emp_count FROM employees;
emp_count
26
```

```
SQL> SELECT department_name, COUNT(*) AS headcount
FROM employee_directory
GROUP BY department_name
ORDER BY headcount DESC;
department_name headcount
Engineering      8
Design           5
Marketing        4
Sales            4
People Ops       3
Finance          2
```

```
SQL> SELECT full_name, department_name, manager_name
FROM employee_directory
LIMIT 5;
full_name      department_name manager_name
Sam Okafor     Engineering    NULL
Priya Nair     Engineering    Sam Okafor
Elena Petrova  Design        Sam Okafor
Grace Adeyemi  Marketing     Sam Okafor
Isabella Rossi Sales          Sam Okafor
```

7. Results

Six departments and twenty-six employees load successfully. Foreign keys prevent orphan department_id values. The view returns each employee with department and manager names for browsing and SQL practice.

8. Conclusion

Normalised tables plus a view provide a maintainable employee directory. Primary and foreign keys preserve referential integrity; the view simplifies read-only reporting queries used in the web compiler.

Ex – 2

CREATE & ALTER TABLE

1. Aim

To create an employees table with attributes empID, empName, department and salary, insert sample records, and then rename the column department to faculty using ALTER TABLE.

2. Theory

2.1 CREATE TABLE

A relational table is defined by a name, a list of attributes, and constraints that protect the data. PRIMARY KEY makes empID unique and not null. NOT NULL rejects missing names and faculty values. CHECK (salary > 0) rejects non-positive salaries. NUMERIC(10, 2) stores pay as a decimal with two fractional digits.

2.2 ALTER TABLE RENAME COLUMN

In SQLite and MySQL 8+, ALTER TABLE ... RENAME COLUMN changes a column's name without rewriting the stored values. After RENAME COLUMN department TO faculty, every row still holds the same school name; only the heading of that column becomes faculty. This is a schema change, not an UPDATE. Microsoft SQL Server does not accept that syntax; the equivalent is EXEC sp_rename 'employees.department', 'faculty', 'COLUMN' (see employees.sqlserver.sql).

3. Schema

The table is created with four attributes. After the rename, the third attribute is faculty instead of department.

Attribute	Type	Constraint	Role
empID	INTEGER	PRIMARY KEY	Employee identifier
empName	TEXT	NOT NULL	Employee name
department → faculty	TEXT	NOT NULL	School / faculty name
salary	NUMERIC(10,2)	NOT NULL, CHECK > 0	Monthly salary

Sample faculty values include **School of Engineering and Technology** (the faculty named on this report's cover). The cover also lists **Department of CSE** as the student's home department; that heading is letterhead, not a fifth column in the table.

4. Procedure

1. Drop employees if it already exists, so the script can be re-run.
2. Create the table with empID, empName, department and salary.
3. Insert six sample employees.
4. Display all rows and the column list with PRAGMA table_info.
5. Rename department to faculty.
6. Display all rows and the column list again to confirm that only the name changed.

5. Source Code

```
-- =====
-- DBMS Lab · 26-08-2026
-- Employees table: CREATE, seed, then RENAME department → faculty
```

```
-- SQLite (sql.js compiler / DB Browser / sqlite3)
-- For the class SQL Server, run employees.sqlserver.sql instead.
-- =====

DROP TABLE IF EXISTS employees;

-- -----
-- 1. Create the employees table
-- -----
CREATE TABLE employees (
    empID      INTEGER PRIMARY KEY,
    empName    TEXT NOT NULL,
    department TEXT NOT NULL,
    salary     NUMERIC(10, 2) NOT NULL CHECK (salary > 0)
);

-- -----
-- 2. Insert sample rows (department holds the faculty/school name)
-- -----
INSERT INTO employees (empID, empName, department, salary) VALUES
    (1, 'Ananya Sharma', 'School of Engineering and Technology', 92000.00),
    (2, 'Rohan Mehta', 'School of Management', 78000.00),
    (3, 'Priya Nair', 'School of Law', 85000.00),
    (4, 'Vikram Singh', 'School of Agricultural Sciences', 76000.00),
    (5, 'Fatima Khan', 'School of Engineering and Technology', 88000.00),
    (6, 'Arjun Patel', 'School of Medical Sciences', 95000.00);

-- -----
-- 3. Verify the table BEFORE the rename
-- -----
SELECT * FROM employees;
PRAGMA table_info(employees);

-- -----
-- 4. Rename the column department → faculty
-- Row values are unchanged; only the column name changes.
-- -----
ALTER TABLE employees RENAME COLUMN department TO faculty;

-- -----
-- 5. Verify the table AFTER the rename
-- -----
SELECT * FROM employees;
PRAGMA table_info(employees);
```

6. Output

The script was executed in SQLite. PRAGMA table_info columns are cid (column index), name, type, notnull, dflt_value and pk.

```
SQL> DROP TABLE IF EXISTS employees;
-- OK

SQL> CREATE TABLE employees (
    empID      INTEGER PRIMARY KEY,
    empName    TEXT NOT NULL,
    department TEXT NOT NULL,
    salary     NUMERIC(10, 2) NOT NULL CHECK (salary > 0)
);
-- OK

SQL> INSERT INTO employees (empID, empName, department, salary) VALUES
    (1, 'Ananya Sharma', 'School of Engineering and Technology', 92000.00),
    (2, 'Rohan Mehta', 'School of Management', 78000.00),
    (3, 'Priya Nair', 'School of Law', 85000.00),
    (4, 'Vikram Singh', 'School of Agricultural Sciences', 76000.00),
    (5, 'Fatima Khan', 'School of Engineering and Technology', 88000.00),
    (6, 'Arjun Patel', 'School of Medical Sciences', 95000.00);
-- 6 row(s) inserted

SQL> SELECT * FROM employees;
empID empName      department                                salary
-----
1     Ananya Sharma School of Engineering and Technology      92000
2     Rohan Mehta   School of Management                     78000
3     Priya Nair    School of Law                           85000
4     Vikram Singh  School of Agricultural Sciences          76000
5     Fatima Khan   School of Engineering and Technology     88000
6     Arjun Patel   School of Medical Sciences              95000

SQL> PRAGMA table_info(employees);
cid  name      type      notnull  dflt_value  pk
----
0    empID     INTEGER   0        0           1
```

```

1  empName  TEXT          1          0
2  department TEXT        1          0
3  salary   NUMERIC(10, 2) 1          0

SQL> ALTER TABLE employees RENAME COLUMN department TO faculty;
-- OK

SQL> SELECT * FROM employees;
empID  empName      faculty                                     salary
-----
1      Ananya Sharma School of Engineering and Technology 92000
2      Rohan Mehta   School of Management                78000
3      Priya Nair    School of Law                       85000
4      Vikram Singh  School of Agricultural Sciences      76000
5      Fatima Khan   School of Engineering and Technology 88000
6      Arjun Patel   School of Medical Sciences           95000

SQL> PRAGMA table_info(employees);
cid  name      type      notnull  dflt_value  pk
-----
0    empID     INTEGER   0        1           1
1    empName   TEXT      1        0           0
2    faculty   TEXT      1        0           0
3    salary    NUMERIC(10, 2) 1        0           0

```

7. Results

Stage	Third column	Row count	Values changed?
Before ALTER	department	6	No — original schema
After ALTER	faculty	6	No — names only

All six employees remain. School names such as School of Engineering and Technology are unchanged. The schema listing is the proof: `cid = 2` is named `department` before the statement and `faculty` afterwards.

8. Conclusion

An employees relation can be created with a primary key, not-null attributes and a salary check, then adjusted with `ALTER TABLE ... RENAME COLUMN`. Renaming `department` to `faculty` updates the schema only. The stored values do not change, which is the correct way to relabel an attribute without rewriting the table.

Ex – 3

SQL Joins

1. Aim

To combine rows from related tables using `CROSS JOIN`, `NATURAL JOIN`, `INNER JOIN`, `LEFT OUTER JOIN`, `RIGHT OUTER JOIN`, and `SELF JOIN`; to compare how each join type treats unmatched rows; and to understand when to use each one.

2. Theory

2.1 CROSS JOIN

A `CROSS JOIN` returns the Cartesian product of two tables: every row from the first table is paired with every row from the second. There is no join condition. With 6 departments and 7 employees the result has $6 \times 7 = 42$ **rows**. Cross joins are rare in everyday queries but useful for generating combinations (e.g. all size–colour pairs) or as a building block for other joins.

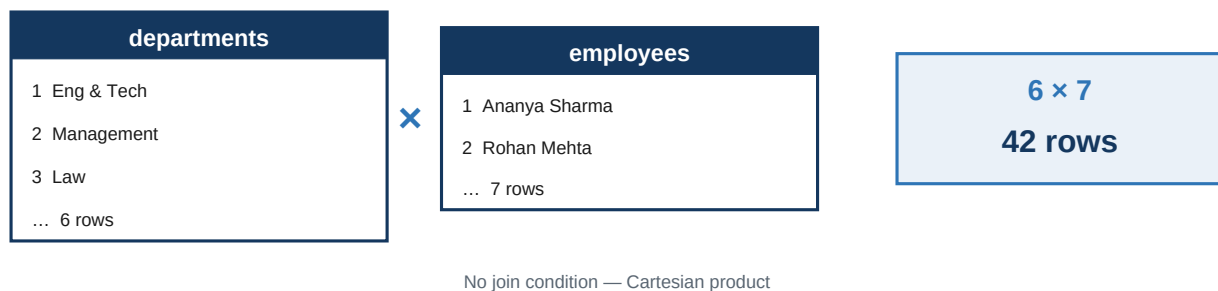


Figure 1 — `CROSS JOIN` pairs every department with every employee (42 combinations).

2.2 NATURAL JOIN

A `NATURAL JOIN` automatically matches rows on every column that appears in both tables with the same name. Here, both tables share `deptID`, so the condition is implicit. Only rows with a matching `deptID` in both tables are returned — the same six rows as `INNER JOIN` when `deptID` is the sole common column. (SQL Server does not support `NATURAL JOIN`; use an explicit `INNER JOIN` instead.)

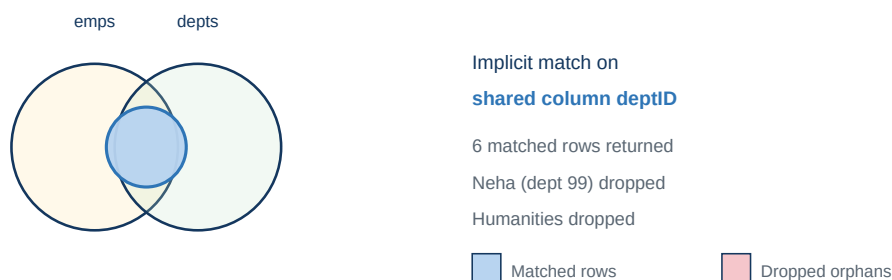


Figure 2 — `NATURAL JOIN` keeps only rows where `deptID` matches in both tables.

2.3 INNER JOIN

An **INNER JOIN** returns only rows where the explicit join condition is satisfied. Unmatched rows from either table are discarded. It is the most common join: use it when you only want records that have a partner on both sides.

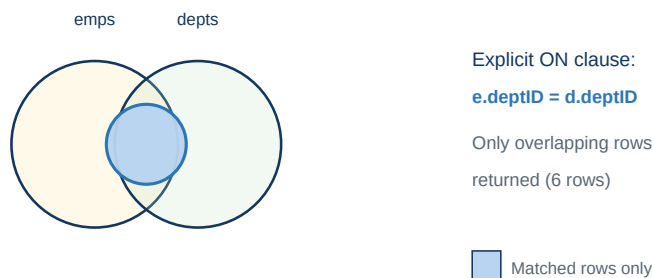


Figure 3 — INNER JOIN returns only the overlapping region (6 matched rows).

2.4 LEFT OUTER JOIN

A **LEFT OUTER JOIN** keeps every row from the left table. If no match exists in the right table, the right-hand columns are filled with **NULL**. Employee Neha Gupta (deptID 99) has no matching department and therefore appears with a null department name.

employees (left table)	LEFT JOIN result
Ananya dept=1	Ananya Eng & Tech
Rohan dept=2	Rohan Management
Priya dept=3	Priya Law
Neha dept=99 ← no match	Neha NULL

All left rows kept — unmatched right columns print as NULL

Figure 4 — LEFT OUTER JOIN preserves Neha (deptID 99); deptName is NULL.

2.5 RIGHT OUTER JOIN

A **RIGHT OUTER JOIN** keeps every row from the right table. Unmatched left-hand columns become **NULL**. School of Humanities (deptID 6) has no employees and therefore appears with null employee fields. A **RIGHT JOIN** can always be rewritten as a **LEFT JOIN** by swapping the table order.

departments (right table)	RIGHT JOIN result
1 Eng & Tech	Ananya Eng & Tech
2 Management	Rohan Management
3 Law	Priya Law
6 Humanities ← no employees	NULL Humanities

All right rows kept — unmatched left columns print as NULL

Figure 5 — RIGHT OUTER JOIN preserves Humanities (deptID 6); employee columns are NULL.

2.6 SELF JOIN

A SELF JOIN joins a table to itself using two different aliases. It is not a separate SQL keyword — it is an INNER or LEFT JOIN where both sides are the same table. Here, employees e is joined to employees m on e.managerID = m.empID to list each employee alongside their manager's name.

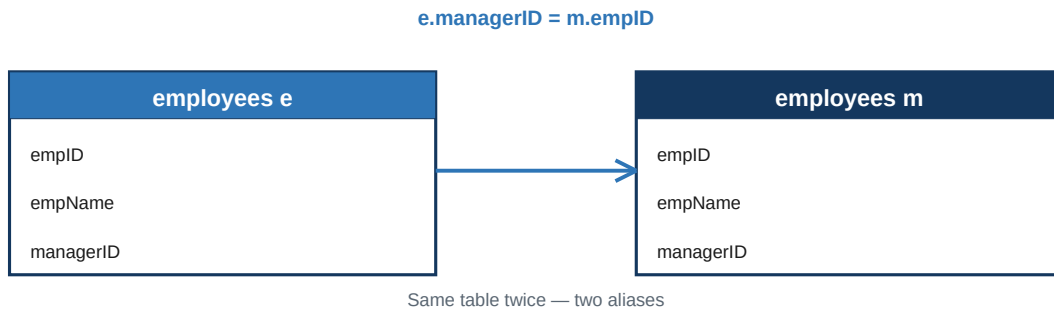


Figure 6 — SELF JOIN: the same table appears twice with different aliases.

2.7 Differences between all join types

Join type	Condition	Unmatched rows	Typical use
CROSS JOIN	None (Cartesian product)	N/A — all combinations	Generate all pairs
NATURAL JOIN	Implicit (shared names)	Dropped from both sides	Quick join on a shared key
INNER JOIN	Explicit ON clause	Dropped from both sides	Only matching records
LEFT OUTER JOIN	Explicit ON clause	Left kept; right NULL	Keep all from the left table
RIGHT OUTER JOIN	Explicit ON clause	Right kept; left NULL	Keep all from the right table
SELF JOIN	Explicit ON (same table)	Depends on INNER / LEFT	Hierarchies, comparisons

Key distinctions: CROSS JOIN multiplies rows with no filter. INNER and NATURAL return only matches. LEFT and RIGHT OUTER preserve one side's orphans. SELF JOIN is a technique (same table, two aliases), not a separate result category — it uses inner or outer join semantics on a single table.

3. Schema

Two tables are linked by deptID. employees also references itself through managerID for the self join (see Figure 6).

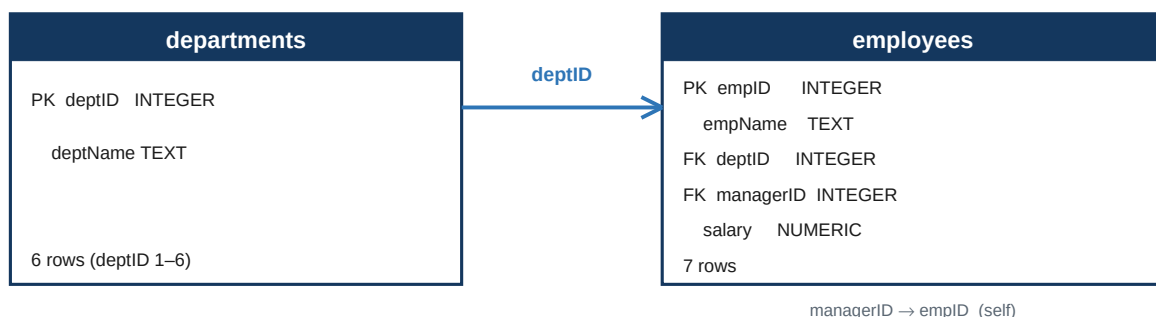


Figure 7 — Schema: employees.deptID references departments.deptID; managerID is a self-reference.

Table	Attribute	Type	Role
departments	deptID	INTEGER / INT	Primary key
departments	deptName	TEXT / VARCHAR	School name
employees	empID	INTEGER / INT	Primary key
employees	empName	TEXT / VARCHAR	Employee name
employees	deptID	INTEGER / INT	FK → departments
employees	managerID	INTEGER / INT	FK → employees (self)
employees	salary	NUMERIC(10,2)	Monthly salary

4. Procedure

1. Create departments and employees (with managerID for self join).
2. Insert six departments and seven employees, including deliberate mismatches for outer joins.
3. Display both base tables.
4. Run `CROSS JOIN` — expect 42 rows (count shown; sample limited to 8).
5. Run `NATURAL JOIN` and `INNER JOIN` — expect six matched rows.
6. Run `LEFT OUTER JOIN` — expect seven rows (one employee without a department).
7. Run `RIGHT OUTER JOIN` — expect seven rows (one department without employees).
8. Run `SELF JOIN` — expect seven rows listing each employee and their manager.

5. Source Code

SQLite script used in the lab compiler (02-09-2026/joins.sql). For the class SQL Server, run joins.sqlserver.sql.

```
-- =====
-- DBMS Lab · 02-09-2026
-- SQL Joins: CROSS, NATURAL, INNER, OUTER (LEFT/RIGHT), SELF
-- SQLite (sql.js compiler / DB Browser / sqlite3)
-- For the class SQL Server, run joins.sqlserver.sql instead.
-- =====

DROP TABLE IF EXISTS employees;
DROP TABLE IF EXISTS departments;

-----
-- 1. Create related tables
--   · deptID links employees → departments
--   · managerID links employees → employees (for SELF JOIN)
-----
CREATE TABLE departments (
    deptID   INTEGER PRIMARY KEY,
    deptName TEXT NOT NULL
);

CREATE TABLE employees (
    empID      INTEGER PRIMARY KEY,
    empName    TEXT NOT NULL,
    deptID     INTEGER NOT NULL,
    managerID  INTEGER,
    salary     NUMERIC(10, 2) NOT NULL CHECK (salary > 0)
);

-----
-- 2. Insert sample rows
--   · deptID 6 (Humanities) has no employees → RIGHT OUTER JOIN
--   · empID 7 points to deptID 99 (missing) → LEFT OUTER JOIN
--   · managerID links form a simple reporting chain → SELF JOIN
-----
INSERT INTO departments (deptID, deptName) VALUES
    (1, 'School of Engineering and Technology'),
    (2, 'School of Management'),
    (3, 'School of Law'),
    (4, 'School of Agricultural Sciences'),
    (5, 'School of Medical Sciences'),
    (6, 'School of Humanities');

INSERT INTO employees (empID, empName, deptID, managerID, salary) VALUES
    (1, 'Ananya Sharma', 1, NULL, 92000.00),
    (2, 'Rohan Mehta', 2, 1, 78000.00),
    (3, 'Priya Nair', 3, 1, 85000.00),
    (4, 'Vikram Singh', 4, 2, 76000.00),
    (5, 'Fatima Khan', 1, 1, 88000.00),
    (6, 'Arjun Patel', 5, 3, 95000.00),
    (7, 'Neha Gupta', 99, 2, 54000.00);

-----
-- 3. Base tables (reference)
-----
-- List all departments
SELECT * FROM departments;
-- List all employees
SELECT * FROM employees;

-----
-- 4. CROSS JOIN
--   Cartesian product: every row of table A paired with every
--   row of table B. No join condition. Here: 6 × 7 = 42 rows.
-----
-- Count cross join rows
SELECT COUNT(*) AS cross_join_row_count
FROM departments
CROSS JOIN employees;

-- Cross join sample pairs
SELECT d.deptName, e.empName
FROM departments AS d
CROSS JOIN employees AS e
ORDER BY d.deptID, e.empID
LIMIT 8;

-----
-- 5. NATURAL JOIN
--   Matches on every column with the same name (here: deptID).
--   Only rows with a matching deptID in BOTH tables appear.
-----
```

```

-- Natural join on deptID
SELECT empID, empName, deptName, salary
FROM employees
NATURAL JOIN departments;

-----
-- 6. INNER JOIN
--   Explicit join condition; same result as NATURAL JOIN here
--   because deptID is the only common column.
-----
-- Inner join employees and departments
SELECT e.empID, e.empName, d.deptName, e.salary
FROM employees AS e
INNER JOIN departments AS d ON e.deptID = d.deptID;

-----
-- 7a. LEFT OUTER JOIN
--   All rows from the LEFT table (employees); unmatched dept
--   columns are NULL (empID 7 → deptID 99 has no department).
-----
-- Left join – keep all employees
SELECT e.empID, e.empName, e.deptID, d.deptName, e.salary
FROM employees AS e
LEFT OUTER JOIN departments AS d ON e.deptID = d.deptID;

-----
-- 7b. RIGHT OUTER JOIN
--   All rows from the RIGHT table (departments); unmatched
--   employee columns are NULL (deptID 6 has no employees).
--   Requires SQLite 3.39+; see joins.sqlserver.sql for T-SQL.
-----
-- Right join – keep all departments
SELECT e.empID, e.empName, d.deptID, d.deptName, e.salary
FROM employees AS e
RIGHT OUTER JOIN departments AS d ON e.deptID = d.deptID;

-----
-- 8. SELF JOIN
--   A table joined to itself using different aliases.
--   Here: each employee matched to their manager (also in
--   employees). Ananya has no manager → manager name is NULL.
-----
-- Self join – employee to manager
SELECT
    e.empName AS employee,
    m.empName AS manager
FROM employees AS e
LEFT JOIN employees AS m ON e.managerID = m.empID
ORDER BY e.empID;

```

6. Output

Each SELECT from the script, executed in SQLite. Unmatched columns print as NULL (highlighted). CREATE and INSERT output is omitted here; the statements are in section 5.

6.1 Base table — departments

```
SQL> SELECT * FROM departments;
```

deptID	deptName
1	School of Engineering and Technology
2	School of Management
3	School of Law
4	School of Agricultural Sciences
5	School of Medical Sciences
6	School of Humanities

6.2 Base table — employees

```
SQL> SELECT * FROM employees;
```

empID	empName	deptID	managerID	salary
1	Ananya Sharma	1	NULL	92000
2	Rohan Mehta	2	1	78000
3	Priya Nair	3	1	85000
4	Vikram Singh	4	2	76000
5	Fatima Khan	1	1	88000
6	Arjun Patel	5	3	95000
7	Neha Gupta	99	2	54000

6.3 CROSS JOIN — row count

```
SQL> SELECT COUNT(*) AS cross_join_row_count
FROM departments
CROSS JOIN employees;
```

cross_join_row_count
42

6.4 CROSS JOIN — first 8 pairs

```
SQL> SELECT d.deptName, e.empName
      FROM departments AS d
      CROSS JOIN employees AS e
      ORDER BY d.deptID, e.empID
      LIMIT 8;
```

deptName	empName
School of Engineering and Technology	Ananya Sharma
School of Engineering and Technology	Rohan Mehta
School of Engineering and Technology	Priya Nair
School of Engineering and Technology	Vikram Singh
School of Engineering and Technology	Fatima Khan
School of Engineering and Technology	Arjun Patel
School of Engineering and Technology	Neha Gupta
School of Management	Ananya Sharma

6.5 NATURAL JOIN

```
SQL> SELECT empID, empName, deptName, salary
      FROM employees
      NATURAL JOIN departments;
```

empID	empName	deptName	salary
1	Ananya Sharma	School of Engineering and Technology	92000
2	Rohan Mehta	School of Management	78000
3	Priya Nair	School of Law	85000
4	Vikram Singh	School of Agricultural Sciences	76000
5	Fatima Khan	School of Engineering and Technology	88000
6	Arjun Patel	School of Medical Sciences	95000

6.6 INNER JOIN

```
SQL> SELECT e.empID, e.empName, d.deptName, e.salary
      FROM employees AS e
      INNER JOIN departments AS d ON e.deptID = d.deptID;
```

empID	empName	deptName	salary
1	Ananya Sharma	School of Engineering and Technology	92000
2	Rohan Mehta	School of Management	78000
3	Priya Nair	School of Law	85000
4	Vikram Singh	School of Agricultural Sciences	76000
5	Fatima Khan	School of Engineering and Technology	88000
6	Arjun Patel	School of Medical Sciences	95000

6.7 LEFT OUTER JOIN

```
SQL> SELECT e.empID, e.empName, e.deptID, d.deptName, e.salary
FROM employees AS e
LEFT OUTER JOIN departments AS d ON e.deptID = d.deptID;
```

empID	empName	deptID	deptName	salary
1	Ananya Sharma	1	School of Engineering and Technology	92000
2	Rohan Mehta	2	School of Management	78000
3	Priya Nair	3	School of Law	85000
4	Vikram Singh	4	School of Agricultural Sciences	76000
5	Fatima Khan	1	School of Engineering and Technology	88000
6	Arjun Patel	5	School of Medical Sciences	95000
7	Neha Gupta	99	NULL	54000

6.8 RIGHT OUTER JOIN

```
SQL> SELECT e.empID, e.empName, d.deptID, d.deptName, e.salary
FROM employees AS e
RIGHT OUTER JOIN departments AS d ON e.deptID = d.deptID;
```

empID	empName	deptID	deptName	salary
1	Ananya Sharma	1	School of Engineering and Technology	92000
2	Rohan Mehta	2	School of Management	78000
3	Priya Nair	3	School of Law	85000
4	Vikram Singh	4	School of Agricultural Sciences	76000
5	Fatima Khan	1	School of Engineering and Technology	88000
6	Arjun Patel	5	School of Medical Sciences	95000
NULL	NULL	6	School of Humanities	NULL

6.9 SELF JOIN — employee and manager

```
SQL> SELECT
    e.empName AS employee,
    m.empName AS manager
FROM employees AS e
LEFT JOIN employees AS m ON e.managerID = m.empID
ORDER BY e.empID;
```

employee	manager
Ananya Sharma	NULL
Rohan Mehta	Ananya Sharma
Priya Nair	Ananya Sharma
Vikram Singh	Rohan Mehta
Fatima Khan	Ananya Sharma
Arjun Patel	Priya Nair
Neha Gupta	Rohan Mehta

7. Results

Row counts returned by each join on this dataset:

Join type	Rows	Unmatched handling
CROSS JOIN	42	No condition — every dept × every employee
NATURAL JOIN	6	Drops non-matching rows from both sides
INNER JOIN	6	Same six rows as NATURAL JOIN here
LEFT OUTER JOIN	7	Keeps Neha Gupta (deptID 99); deptName is NULL
RIGHT OUTER JOIN	7	Keeps Humanities (deptID 6); employee cols are NULL
SELF JOIN	7	All employees; Ananya has NULL manager (top of chain)

CROSS JOIN produces the largest result because it ignores relationships entirely. INNER JOIN and NATURAL JOIN return only the six employees whose deptID exists in departments. Outer joins add the orphan employee (left) or orphan department (right). SELF JOIN does not change row count here because a LEFT JOIN keeps every employee and simply leaves the manager column null for Ananya Sharma, who has no manager.

8. Conclusion

SQL joins answer different questions about related data. CROSS JOIN asks “every combination?” INNER JOIN asks “only matches?” Outer joins ask “keep everyone from this side even without a match?” SELF JOIN asks “how do rows in this table relate to other rows in the same table?” Choosing the right join depends on whether you need all combinations, only matches, or orphan rows from a specific side.

Ex – 4

SELECT Queries

1. Aim

To answer the lab questions below: create an employee table with the given attributes, insert 10 rows, and run the listed SELECT queries (DISTINCT, WHERE, BETWEEN, IN, ORDER BY).

2. Theory

2.1 SELECT

SELECT reads rows from a relation. SELECT * returns every column; naming columns returns only those attributes. A query without a WHERE clause returns every row that currently exists in the table.

2.2 DISTINCT

SELECT DISTINCT removes duplicate values from the result. Question 3 on the lab sheet asks for a “distant” department number; that is DISTINCT. SELECT department_no lists a department once per employee (question 4); SELECT DISTINCT department_no lists each department number only once.

2.3 WHERE

The WHERE clause keeps rows that satisfy a condition. Comparison operators such as > and <> filter numbers and strings. OR keeps a row if either predicate is true (city is Delhi or Mumbai). City values in this lab are stored in title case and matched with the same casing so the filters work in SQLite without extra case-conversion functions.

2.4 BETWEEN

BETWEEN low AND high is an inclusive range: the endpoints are part of the result. NOT BETWEEN is the complement — salaries strictly below 30000 or strictly above 50000. It is equivalent to salary < 30000 OR salary > 50000 when the range is 30000 to 50000.

2.5 IN

IN (...) tests membership in a list. city IN ('Delhi', 'Mumbai', 'Chennai', 'Bangalore') keeps those four cities. NOT IN keeps every city that is not in the list — here Bangalore, Hyderabad, and Pune remain when Mumbai, Delhi, and Chennai are excluded.

2.6 ORDER BY

ORDER BY sorts the result. ASC (the default) lists names A→Z; DESC lists them Z→A. Sorting does not change stored rows; it only changes the order of the result set.

3. Schema

One table, employee, holds eight attributes. sr_no is the employee number (primary key). manager stores the manager's name, or NULL for the top manager.

Attribute	Type	Constraint	Role
sr_no	INTEGER	PRIMARY KEY	Employee number
employee_name	TEXT	NOT NULL	Employee name
job	TEXT	NOT NULL	Job title
manager	TEXT	NULL allowed	Manager name
hire_date	DATE	NOT NULL	Date of joining
salary	NUMERIC(10,2)	NOT NULL, CHECK > 0	Monthly salary
department_no	INTEGER	NOT NULL	Department number
city	TEXT	NOT NULL	Work city

4. Questions

The experiment questions are reproduced as written on the lab sheet (two items numbered 1). Question 3 writes “distant”; the SQL keyword is DISTINCT. Employee number is stored as sr_no.

Q. No.	Question (as given)
1	create a table employee with the following attributes: sr no, employee name, job, manager, hire date, salary, department no, city
1	insert 10 rows in the above mentioned table
2	select all the details from the employee table
3	select only distant department number from employee table
4	select all department numbers from employee table
5	select employee name, salary where salary is >30000 rs from employee table
6	select employee name, hire date where city is not delhi from employee table
7	select employee name, hire date where city is either delhi or mumbai from employee table
8	select employee number, employee name where salary is between 30000 rs and 50000 rs
9	select employee number, employee name where salary is not between 30000 and 50000 from employee table
10	select all the details where city is among the following: delhi, mumbai, chennai, Bangalore
11	select all the details where city is not mumbai, delhi or chennai from employee table
12	select all the details from the employee table and order by employee names in descending order.
13	display the list of employees in ascending order from employee table.

5. Procedure

Drop employee if it already exists so the script can be re-run. Then answer each lab question in order. City values are stored in title case (Delhi, Mumbai, ...) and matched with that same casing. The SQL for each question:

Q. No.	SQL solution
1	<pre>CREATE TABLE employee (sr_no INTEGER PRIMARY KEY, employee_name TEXT NOT NULL, job TEXT NOT NULL, manager TEXT, hire_date DATE NOT NULL, salary NUMERIC(10, 2) NOT NULL CHECK (salary > 0), department_no INTEGER NOT NULL, city TEXT NOT NULL);</pre>
1	<pre>INSERT INTO employee (sr_no, employee_name, job, manager, hire_date, salary, department_no, city) VALUES (1, 'Rajesh Kumar', 'General Manager', NULL, '2018-03-12', 85000.00, 10, 'Delhi'), (2, 'Ananya Sharma', 'Software Engineer', 'Rajesh Kumar', '2021-07-01', 45000.00, 20, 'Bangalore'), (3, 'Rohan Mehta', 'Accountant', 'Rajesh Kumar', '2020-01-15', 28000.00, 30, 'Mumbai'), (4, 'Priya Nair', 'HR Executive', 'Rajesh Kumar', '2022-04-20', 32000.00, 40, 'Chennai'), (5, 'Vikram Singh', 'Sales Manager', 'Rajesh Kumar', '2019-11-08', 52000.00, 50, 'Hyderabad'), (6, 'Fatima Khan', 'Software Engineer', 'Ananya Sharma', '2023-02-14', 38000.00, 20, 'Bangalore'), (7, 'Arjun Patel', 'Clerk', 'Rohan Mehta', '2024-06-01', 18000.00, 30, 'Pune'), (8, 'Neha Gupta', 'Marketing Analyst', 'Rajesh Kumar', '2021-09-10', 35000.00, 60, 'Delhi'), (9, 'Karan Iyer', 'Database Admin', 'Ananya Sharma', '2020-12-05', 48000.00, 20, 'Mumbai'), (10, 'Meera Joshi', 'Receptionist', 'Priya Nair', '2023-08-22', 22000.00, 40, 'Chennai');</pre>
2	<pre>SELECT * FROM employee;</pre>
3	<pre>SELECT DISTINCT department_no FROM employee;</pre>
4	<pre>SELECT department_no FROM employee;</pre>
5	<pre>SELECT employee_name, salary FROM employee WHERE salary > 30000;</pre>
6	<pre>SELECT employee_name, hire_date FROM employee WHERE city <> 'Delhi';</pre>
7	<pre>SELECT employee_name, hire_date FROM employee WHERE city = 'Delhi' OR city = 'Mumbai';</pre>
8	<pre>SELECT sr_no, employee_name FROM employee WHERE salary BETWEEN 30000 AND 50000;</pre>
9	<pre>SELECT sr_no, employee_name FROM employee WHERE salary NOT BETWEEN 30000 AND 50000;</pre>
10	<pre>SELECT * FROM employee WHERE city IN ('Delhi', 'Mumbai', 'Chennai', 'Bangalore');</pre>
11	<pre>SELECT * FROM employee WHERE city NOT IN ('Mumbai', 'Delhi', 'Chennai');</pre>
12	<pre>SELECT * FROM employee ORDER BY employee_name DESC;</pre>
13	<pre>SELECT * FROM employee ORDER BY employee_name ASC;</pre>

6. Source Code

SQLite script used in the lab compiler (09-09-2026/employee.sql). For the class SQL Server, run employee.sqlserver.sql.

```
-- =====
-- DBMS Lab · 09-09-2026
-- Employee table: CREATE, seed 10 rows, then SELECT queries
-- DISTINCT, WHERE, BETWEEN, IN, ORDER BY
-- SQLite (sql.js compiler / DB Browser / sqlite3)
-- For the class SQL Server, run employee.sqlserver.sql instead.
--
```

```

-- Lab questions (wording as given; "distant" = DISTINCT)
-- 1. create a table employee with the following attributes:
--    sr no, employee name, job, manager, hire date, salary,
--    department no, city
-- 1. insert 10 rows in the above mentioned table
-- 2. select all the details from the employee table
-- 3. select only distant department number from employee table
-- 4. select all department numbers from employee table
-- 5. select employee name, salary where salary is >30000 rs
--    from employee table
-- 6. select employee name, hire date where city is not delhi
--    from employee table
-- 7. select employee name, hire date where city is either
--    delhi or mumbai from employee table
-- 8. select employee number, employee name where salary is
--    between 30000 rs and 50000 rs
-- 9. select employee number, employee name where salary is
--    not between 30000 and 50000 from employee table
-- 10. select all the details where city is among the following:
--    delhi, mumbai, chennai, Bangalore
-- 11. select all the details where city is not mumbai, delhi
--    or chennai from employee table
-- 12. select all the details from the employee table and
--    order by employee names in descending order.
-- 13. display the list of employees in ascending order from
--    employee table.
-- =====

DROP TABLE IF EXISTS employee;

-- -----
-- 1. create a table employee with the following attributes:
--    sr no, employee name, job, manager, hire date, salary,
--    department no, city
-- -----
CREATE TABLE employee (
    sr_no          INTEGER PRIMARY KEY,
    employee_name  TEXT NOT NULL,
    job            TEXT NOT NULL,
    manager        TEXT,
    hire_date      DATE NOT NULL,
    salary         NUMERIC(10, 2) NOT NULL CHECK (salary > 0),
    department_no  INTEGER NOT NULL,
    city           TEXT NOT NULL
);

-- -----
-- 1. insert 10 rows in the above mentioned table
--    • salaries below, inside, and above 30000-50000
--    • cities: Delhi, Mumbai, Chennai, Bangalore, Hyderabad, Pune
--    • mixed department numbers (some duplicates)
--    • Rajesh Kumar is the top manager (manager IS NULL)
-- -----
INSERT INTO employee
(sr_no, employee_name, job, manager, hire_date, salary, department_no, city)
VALUES
(1, 'Rajesh Kumar', 'General Manager', NULL, '2018-03-12', 85000.00, 10, 'Delhi'),
(2, 'Ananya Sharma', 'Software Engineer', 'Rajesh Kumar', '2021-07-01', 45000.00, 20, 'Bangalore'),
(3, 'Rohan Mehta', 'Accountant', 'Rajesh Kumar', '2020-01-15', 28000.00, 30, 'Mumbai'),
(4, 'Priya Nair', 'HR Executive', 'Rajesh Kumar', '2022-04-20', 32000.00, 40, 'Chennai'),
(5, 'Vikram Singh', 'Sales Manager', 'Rajesh Kumar', '2019-11-08', 52000.00, 50, 'Hyderabad'),
(6, 'Fatima Khan', 'Software Engineer', 'Ananya Sharma', '2023-02-14', 38000.00, 20, 'Bangalore'),
(7, 'Arjun Patel', 'Clerk', 'Rohan Mehta', '2024-06-01', 18000.00, 30, 'Pune'),
(8, 'Neha Gupta', 'Marketing Analyst', 'Rajesh Kumar', '2021-09-10', 35000.00, 60, 'Delhi'),
(9, 'Karan Iyer', 'Database Admin', 'Ananya Sharma', '2020-12-05', 48000.00, 20, 'Mumbai'),
(10, 'Meera Joshi', 'Receptionist', 'Priya Nair', '2023-08-22', 22000.00, 40, 'Chennai');

-- -----
-- 2. select all the details from the employee table
-- -----
SELECT * FROM employee;

-- -----
-- 3. select only distant department number from employee table
--    ("distant" on the lab sheet = DISTINCT)
-- -----
-- Distant (DISTINCT) department number
SELECT DISTINCT department_no FROM employee;

-- -----
-- 4. select all department numbers from employee table
-- -----
-- All department numbers
SELECT department_no FROM employee;

-- -----
-- 5. select employee name, salary where salary is >30000 rs
--    from employee table
-- -----

```

```

-- Salary > 30000 rs
SELECT employee_name, salary
FROM employee
WHERE salary > 30000;

-----
-- 6. select employee name, hire date where city is not delhi
--    from employee table
-----
-- City is not delhi
SELECT employee_name, hire_date
FROM employee
WHERE city <> 'Delhi';

-----
-- 7. select employee name, hire date where city is either
--    delhi or mumbai from employee table
-----
-- City is either delhi or mumbai
SELECT employee_name, hire_date
FROM employee
WHERE city = 'Delhi' OR city = 'Mumbai';

-----
-- 8. select employee number, employee name where salary is
--    between 30000 rs and 50000 rs
-----
-- Salary between 30000 rs and 50000 rs
SELECT sr_no, employee_name
FROM employee
WHERE salary BETWEEN 30000 AND 50000;

-----
-- 9. select employee number, employee name where salary is
--    not between 30000 and 50000 from employee table
-----
-- Salary not between 30000 and 50000
SELECT sr_no, employee_name
FROM employee
WHERE salary NOT BETWEEN 30000 AND 50000;

-----
-- 10. select all the details where city is among the following:
--      delhi, mumbai, chennai, Bangalore
-----
-- City among delhi, mumbai, chennai, Bangalore
SELECT *
FROM employee
WHERE city IN ('Delhi', 'Mumbai', 'Chennai', 'Bangalore');

-----
-- 11. select all the details where city is not mumbai, delhi
--      or chennai from employee table
-----
-- City is not mumbai, delhi or chennai
SELECT *
FROM employee
WHERE city NOT IN ('Mumbai', 'Delhi', 'Chennai');

-----
-- 12. select all the details from the employee table and
--      order by employee names in descending order.
-----
-- Order by employee names descending
SELECT *
FROM employee
ORDER BY employee_name DESC;

-----
-- 13. display the list of employees in ascending order from
--      employee table.
-----
-- Display employees in ascending order
SELECT *
FROM employee
ORDER BY employee_name ASC;

```

7. Output

The script was executed in SQLite in memory. CREATE and INSERT print a status line; each SELECT prints its result set. NULL in the manager column is shown as an empty cell.

```

SQL> DROP TABLE IF EXISTS employee;
-- OK

SQL> CREATE TABLE employee (
    sr_no          INTEGER PRIMARY KEY,
    employee_name  TEXT NOT NULL,

```

```

        job            TEXT NOT NULL,
        manager        TEXT,
        hire_date       DATE NOT NULL,
        salary          NUMERIC(10, 2) NOT NULL CHECK (salary > 0),
        department_no   INTEGER NOT NULL,
        city            TEXT NOT NULL
    );
-- OK

SQL> INSERT INTO employee
(sr_no, employee_name, job, manager, hire_date, salary, department_no, city)
VALUES
(1, 'Rajesh Kumar', 'General Manager', NULL, '2018-03-12', 85000.00,
10, 'Delhi'),
(2, 'Ananya Sharma', 'Software Engineer', 'Rajesh Kumar', '2021-07-01', 45000.00,
20, 'Bangalore'),
(3, 'Rohan Mehta', 'Accountant', 'Rajesh Kumar', '2020-01-15', 28000.00,
30, 'Mumbai'),
(4, 'Priya Nair', 'HR Executive', 'Rajesh Kumar', '2022-04-20', 32000.00,
40, 'Chennai'),
(5, 'Vikram Singh', 'Sales Manager', 'Rajesh Kumar', '2019-11-08', 52000.00,
50, 'Hyderabad'),
(6, 'Fatima Khan', 'Software Engineer', 'Ananya Sharma', '2023-02-14', 38000.00,
20, 'Bangalore'),
(7, 'Arjun Patel', 'Clerk', 'Rohan Mehta', '2024-06-01', 18000.00,
30, 'Pune'),
(8, 'Neha Gupta', 'Marketing Analyst', 'Rajesh Kumar', '2021-09-10', 35000.00,
60, 'Delhi'),
(9, 'Karan Iyer', 'Database Admin', 'Ananya Sharma', '2020-12-05', 48000.00,
20, 'Mumbai'),
(10, 'Meera Joshi', 'Receptionist', 'Priya Nair', '2023-08-22', 22000.00,
40, 'Chennai');
-- 10 row(s) inserted

SQL> SELECT * FROM employee;
sr_no  employee_name  job                manager      hire_date  salary  department_no  city
-----
1      Rajesh Kumar    General Manager
2      Ananya Sharma   Software Engineer  Rajesh Kumar 2021-07-01  45000   20             Bangalore
3      Rohan Mehta     Accountant         Rajesh Kumar 2020-01-15  28000   30             Mumbai
4      Priya Nair      HR Executive       Rajesh Kumar 2022-04-20  32000   40             Chennai
5      Vikram Singh    Sales Manager      Rajesh Kumar 2019-11-08  52000   50             Hyderabad
6      Fatima Khan     Software Engineer  Ananya Sharma 2023-02-14  38000   20             Bangalore
7      Arjun Patel     Clerk             Rohan Mehta  2024-06-01  18000   30             Pune
8      Neha Gupta      Marketing Analyst  Rajesh Kumar 2021-09-10  35000   60             Delhi
9      Karan Iyer      Database Admin     Ananya Sharma 2020-12-05  48000   20             Mumbai
10     Meera Joshi     Receptionist       Priya Nair    2023-08-22  22000   40             Chennai

SQL> SELECT DISTINCT department_no FROM employee;
department_no
-----
10
20
30
40
50
60

SQL> SELECT department_no FROM employee;
department_no
-----
10
20
30
40
50
20
30
60
20
40

SQL> SELECT employee_name, salary
FROM employee
WHERE salary > 30000;
employee_name  salary
-----
Rajesh Kumar   85000
Ananya Sharma  45000
Priya Nair     32000
Vikram Singh   52000
Fatima Khan    38000
Neha Gupta     35000
Karan Iyer     48000

SQL> SELECT employee_name, hire_date
FROM employee
WHERE city <> 'Delhi';
employee_name  hire_date
-----
Ananya Sharma  2021-07-01
Rohan Mehta    2020-01-15
Priya Nair     2022-04-20
Vikram Singh   2019-11-08
Fatima Khan    2023-02-14
Arjun Patel    2024-06-01
Karan Iyer     2020-12-05
Meera Joshi    2023-08-22

SQL> SELECT employee_name, hire_date
FROM employee
WHERE city = 'Delhi' OR city = 'Mumbai';

```

```

employee_name  hire_date
-----
Rajesh Kumar   2018-03-12
Rohan Mehta    2020-01-15
Neha Gupta     2021-09-10
Karan Iyer     2020-12-05

```

```

SQL> SELECT sr_no, employee_name
      FROM employee
      WHERE salary BETWEEN 30000 AND 50000;

```

```

sr_no  employee_name
-----
2      Ananya Sharma
4      Priya Nair
6      Fatima Khan
8      Neha Gupta
9      Karan Iyer

```

```

SQL> SELECT sr_no, employee_name
      FROM employee
      WHERE salary NOT BETWEEN 30000 AND 50000;

```

```

sr_no  employee_name
-----
1      Rajesh Kumar
3      Rohan Mehta
5      Vikram Singh
7      Arjun Patel
10     Meera Joshi

```

```

SQL> SELECT *
      FROM employee
      WHERE city IN ('Delhi', 'Mumbai', 'Chennai', 'Bangalore');

```

sr_no	employee_name	job	manager	hire_date	salary	department_no	city
1	Rajesh Kumar	General Manager		2018-03-12	85000	10	Delhi
2	Ananya Sharma	Software Engineer	Rajesh Kumar	2021-07-01	45000	20	Bangalore
3	Rohan Mehta	Accountant	Rajesh Kumar	2020-01-15	28000	30	Mumbai
4	Priya Nair	HR Executive	Rajesh Kumar	2022-04-20	32000	40	Chennai
6	Fatima Khan	Software Engineer	Ananya Sharma	2023-02-14	38000	20	Bangalore
8	Neha Gupta	Marketing Analyst	Rajesh Kumar	2021-09-10	35000	60	Delhi
9	Karan Iyer	Database Admin	Ananya Sharma	2020-12-05	48000	20	Mumbai
10	Meera Joshi	Receptionist	Priya Nair	2023-08-22	22000	40	Chennai

```

SQL> SELECT *
      FROM employee
      WHERE city NOT IN ('Mumbai', 'Delhi', 'Chennai');

```

sr_no	employee_name	job	manager	hire_date	salary	department_no	city
2	Ananya Sharma	Software Engineer	Rajesh Kumar	2021-07-01	45000	20	Bangalore
5	Vikram Singh	Sales Manager	Rajesh Kumar	2019-11-08	52000	50	Hyderabad
6	Fatima Khan	Software Engineer	Ananya Sharma	2023-02-14	38000	20	Bangalore
7	Arjun Patel	Clerk	Rohan Mehta	2024-06-01	18000	30	Pune

```

SQL> SELECT *
      FROM employee
      ORDER BY employee_name DESC;

```

sr_no	employee_name	job	manager	hire_date	salary	department_no	city
5	Vikram Singh	Sales Manager	Rajesh Kumar	2019-11-08	52000	50	Hyderabad
3	Rohan Mehta	Accountant	Rajesh Kumar	2020-01-15	28000	30	Mumbai
1	Rajesh Kumar	General Manager		2018-03-12	85000	10	Delhi
4	Priya Nair	HR Executive	Rajesh Kumar	2022-04-20	32000	40	Chennai
8	Neha Gupta	Marketing Analyst	Rajesh Kumar	2021-09-10	35000	60	Delhi
10	Meera Joshi	Receptionist	Priya Nair	2023-08-22	22000	40	Chennai
9	Karan Iyer	Database Admin	Ananya Sharma	2020-12-05	48000	20	Mumbai
6	Fatima Khan	Software Engineer	Ananya Sharma	2023-02-14	38000	20	Bangalore
7	Arjun Patel	Clerk	Rohan Mehta	2024-06-01	18000	30	Pune
2	Ananya Sharma	Software Engineer	Rajesh Kumar	2021-07-01	45000	20	Bangalore

```

SQL> SELECT *
      FROM employee
      ORDER BY employee_name ASC;

```

sr_no	employee_name	job	manager	hire_date	salary	department_no	city
2	Ananya Sharma	Software Engineer	Rajesh Kumar	2021-07-01	45000	20	Bangalore
7	Arjun Patel	Clerk	Rohan Mehta	2024-06-01	18000	30	Pune
6	Fatima Khan	Software Engineer	Ananya Sharma	2023-02-14	38000	20	Bangalore
9	Karan Iyer	Database Admin	Ananya Sharma	2020-12-05	48000	20	Mumbai
10	Meera Joshi	Receptionist	Priya Nair	2023-08-22	22000	40	Chennai
8	Neha Gupta	Marketing Analyst	Rajesh Kumar	2021-09-10	35000	60	Delhi
4	Priya Nair	HR Executive	Rajesh Kumar	2022-04-20	32000	40	Chennai
1	Rajesh Kumar	General Manager		2018-03-12	85000	10	Delhi
3	Rohan Mehta	Accountant	Rajesh Kumar	2020-01-15	28000	30	Mumbai
5	Vikram Singh	Sales Manager	Rajesh Kumar	2019-11-08	52000	50	Hyderabad

8. Results

Ten employees load successfully. Each lab question returns a non-empty result on this seed data:

Q. No.	Rows	What the result shows
1 (create)	—	Table employee created with 8 attributes
1 (insert)	10	10 rows inserted
2	10	All details from the employee table
3	6	Distant/DISTINCT department numbers: 10, 20, 30, 40, 50, 60
4	10	All department numbers (dept 20 appears three times)
5	7	employee name, salary where salary is >30000 rs
6	8	employee name, hire date where city is not delhi
7	4	employee name, hire date where city is delhi or mumbai
8	5	employee number, name where salary is between 30000 rs and 50000 rs
9	5	employee number, name where salary is not between 30000 and 50000
10	8	all details where city is delhi, mumbai, chennai, Bangalore
11	4	all details where city is not mumbai, delhi or chennai
12	10	all details ordered by employee names descending
13	10	list of employees in ascending order

Question 3 (DISTINCT) collapses ten department values to six unique numbers; question 4 lists all ten. Questions 8 and 9 partition salaries with no overlap ($5 + 5 = 10$). Question 11 keeps Bangalore, Hyderabad, and Pune — cities that are not mumbai, delhi, or chennai.

9. Conclusion

A single SELECT can project columns, remove duplicates, filter with comparison and set predicates, and sort the result. DISTINCT answers “which unique values exist?” WHERE answers “which rows match this condition?” BETWEEN and IN express ranges and lists clearly. ORDER BY changes presentation only. Together these clauses are the core of everyday retrieval in SQL.